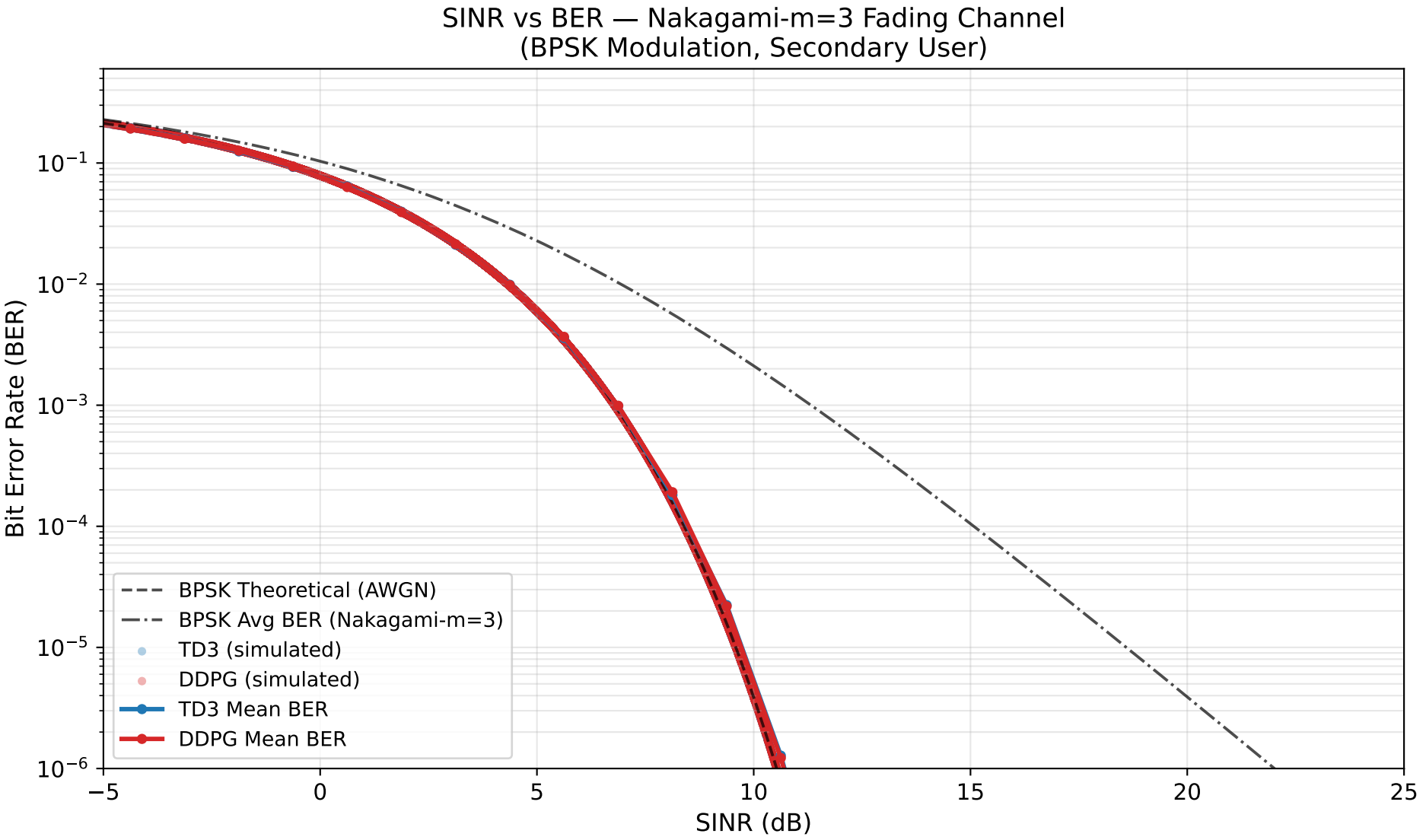
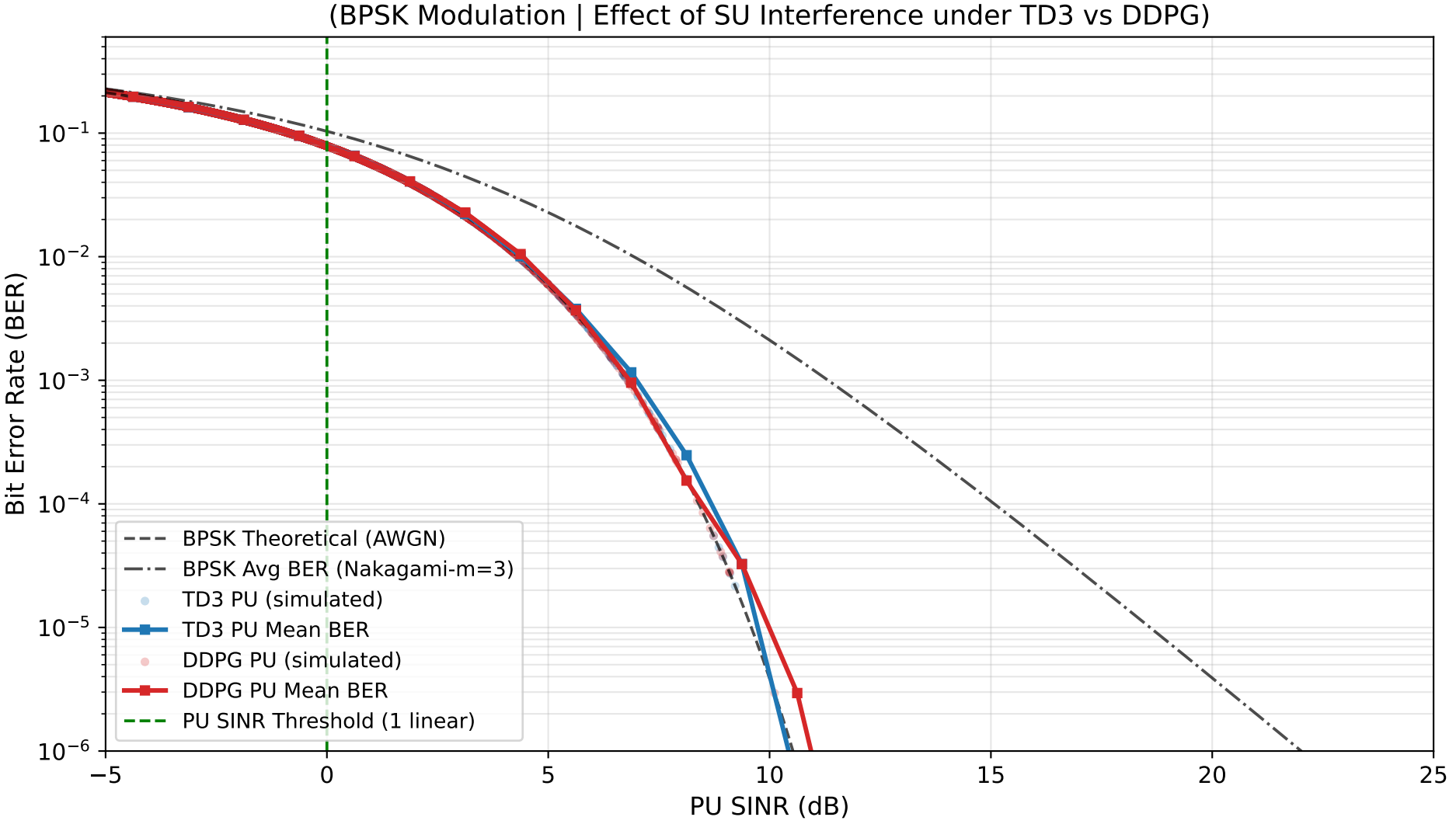
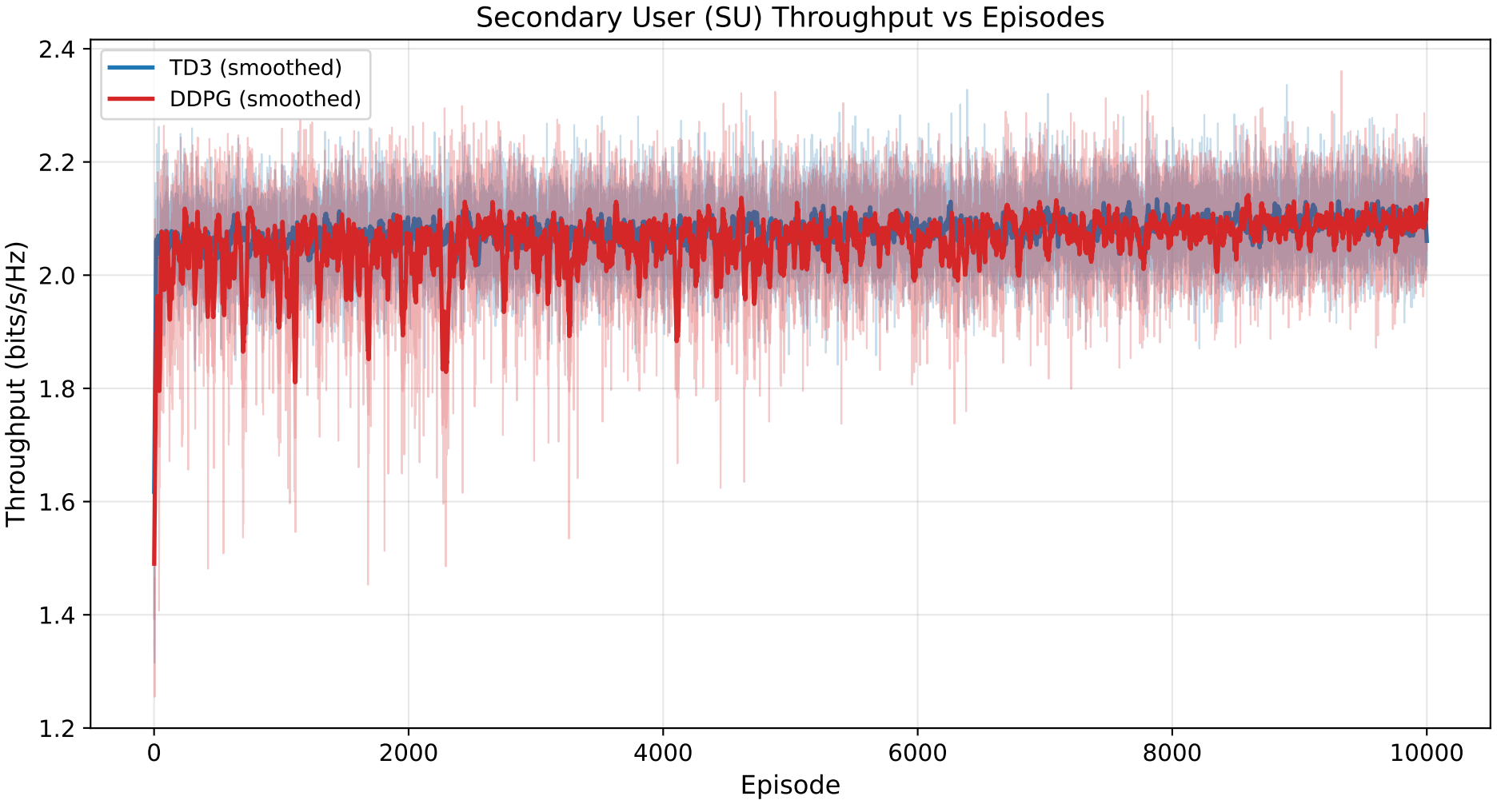
CRN Power Control: TD3 vs DDPG
Nakagami-m Fading Channel Performance Report

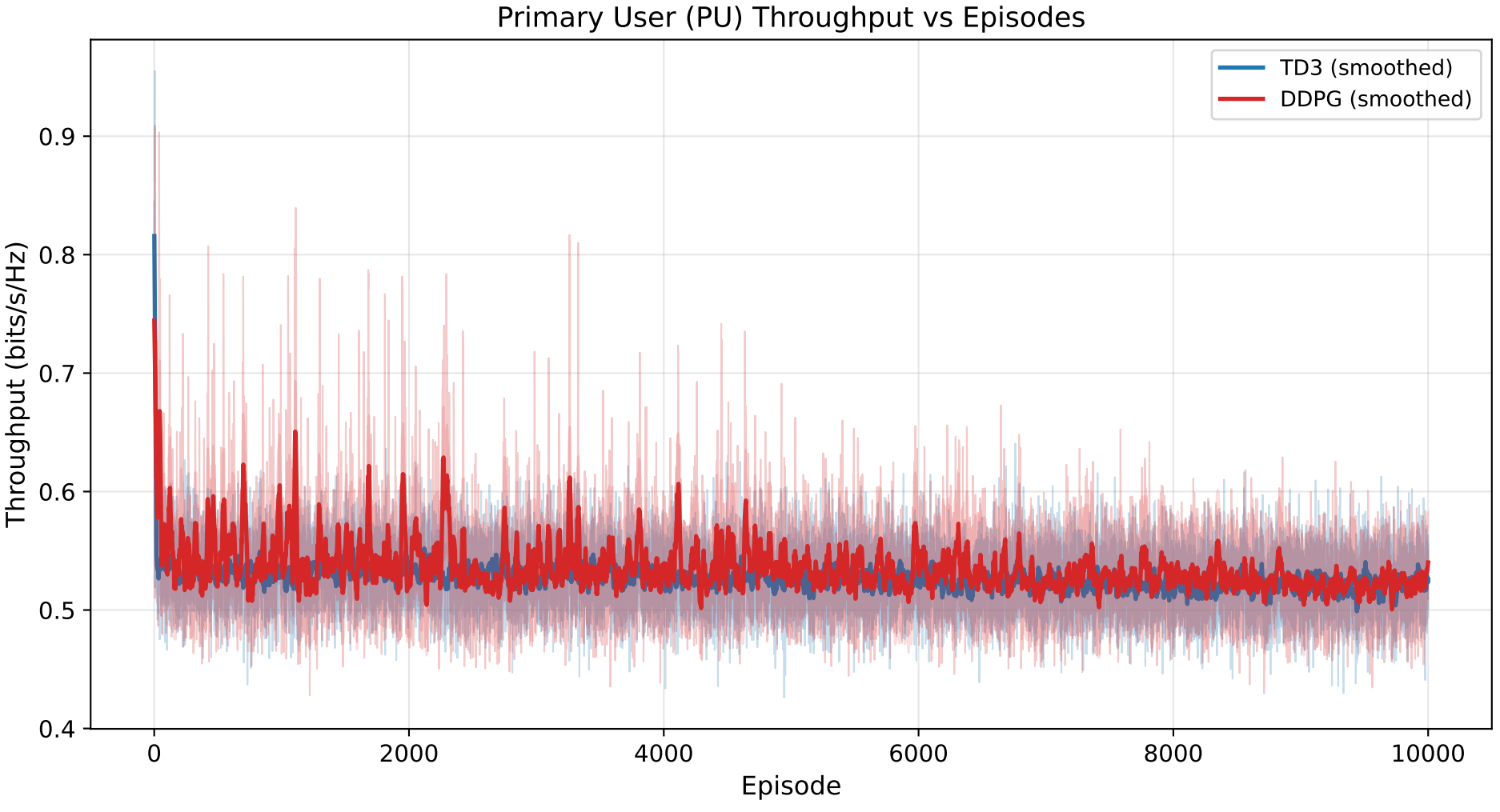
Nakagami-m = 3.0 | Episodes = 10000 | Steps/Episode = 200 | SINR Threshold = 1.0 dB

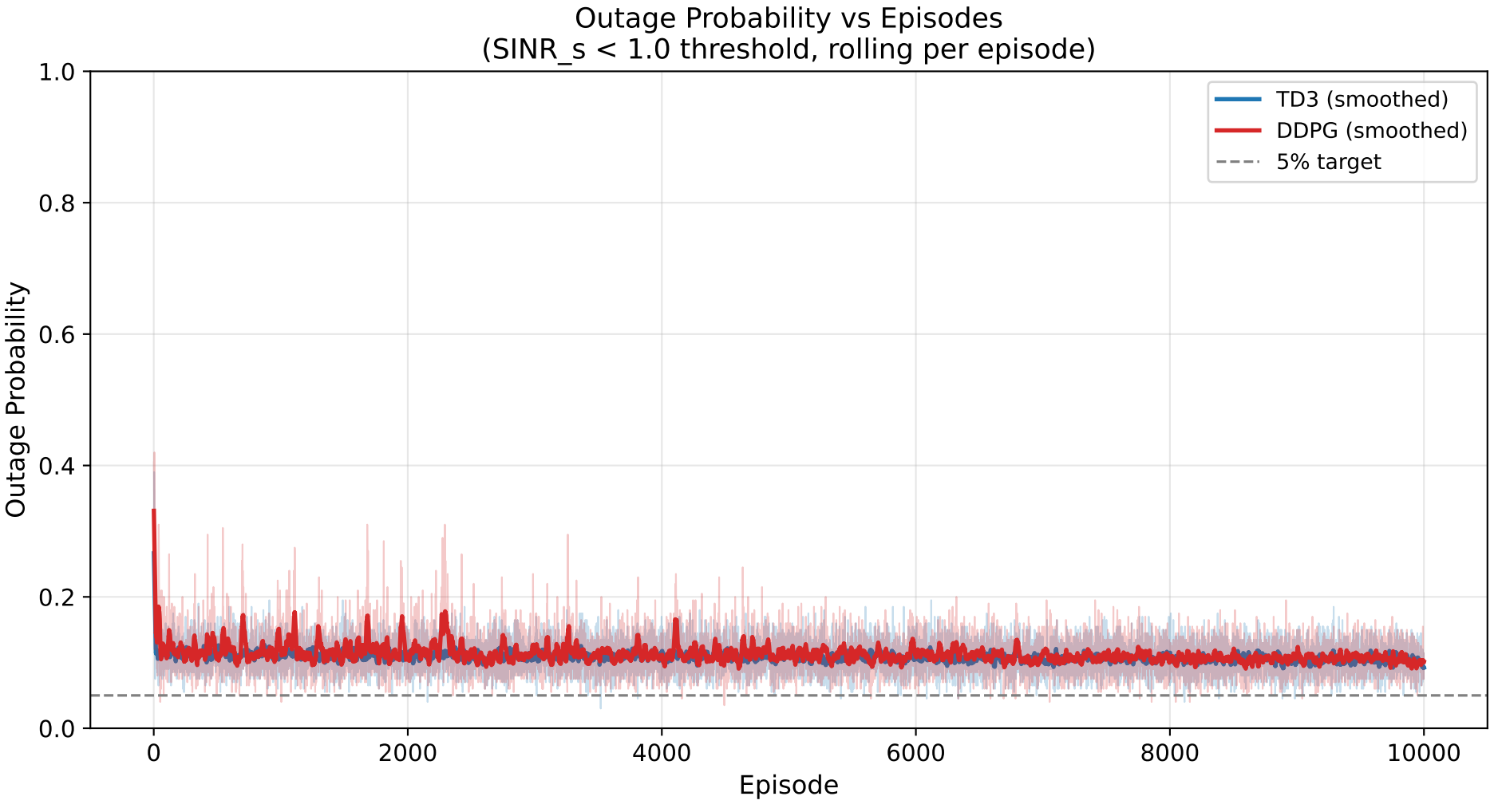
Metric	TD3	DDPG	Winner
Avg Reward (last 100 ep)	247.1264	246.3682	**TD3**
SU Throughput (bits/s/Hz)	2.1043	2.0968	**TD3**
PU Throughput (bits/s/Hz)	0.5255	0.5263	**DDPG**
Outage Probability	0.1027	0.0999	**DDPG**
Average BER	0.026232	0.026127	**DDPG**
Training Time (s)	64466.3	61532.7	—

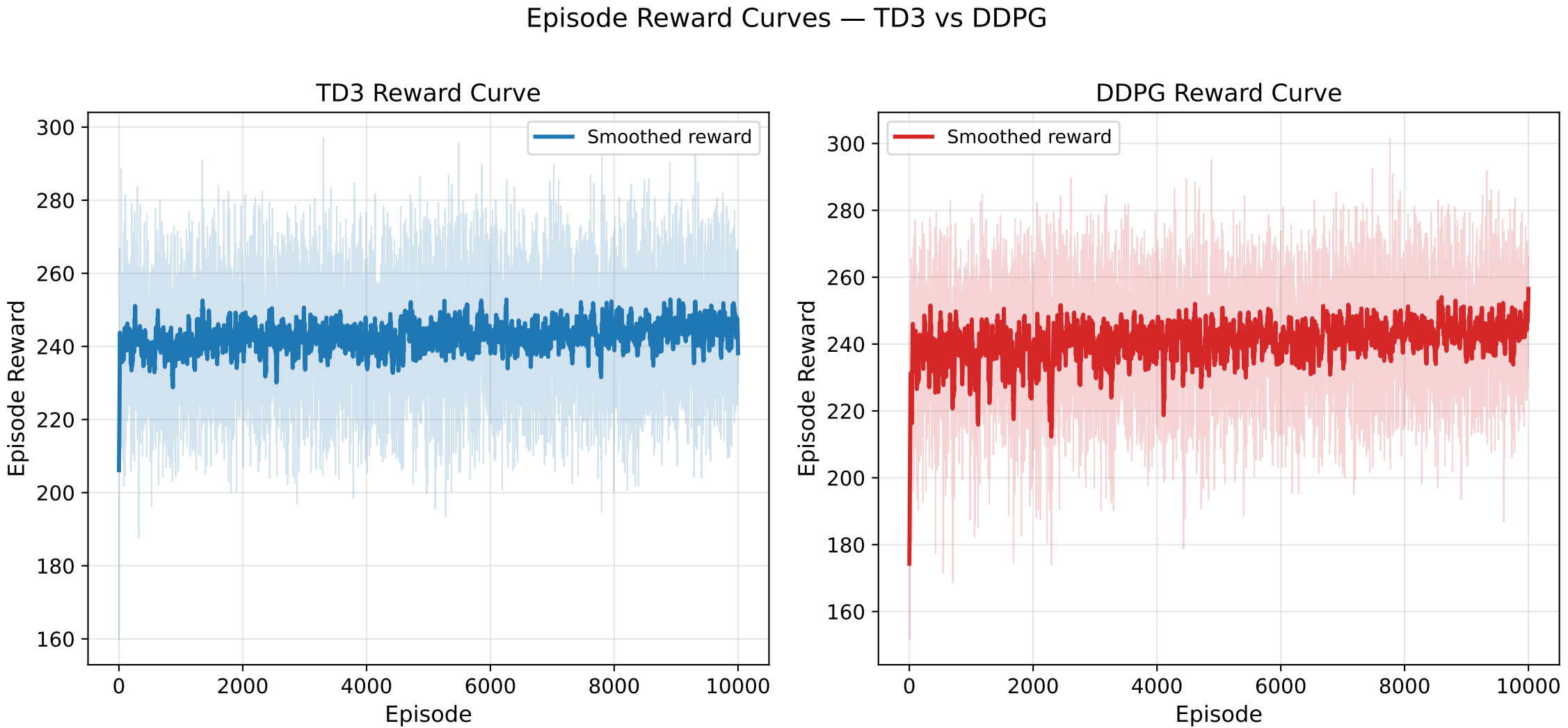

SINR vs BER — Nakagami-m=3 Fading Channel
(BPSK Modulation, Secondary User)
Bit Error Rate (BER)
SINR (dB)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
−5
0
5
10
15
20
25
BPSK Theoretical (AWGN)
BPSK Avg BER (Nakagami-m=3)
TD3 (simulated)
DDPG (simulated)
TD3 Mean BER
DDPG Mean BER

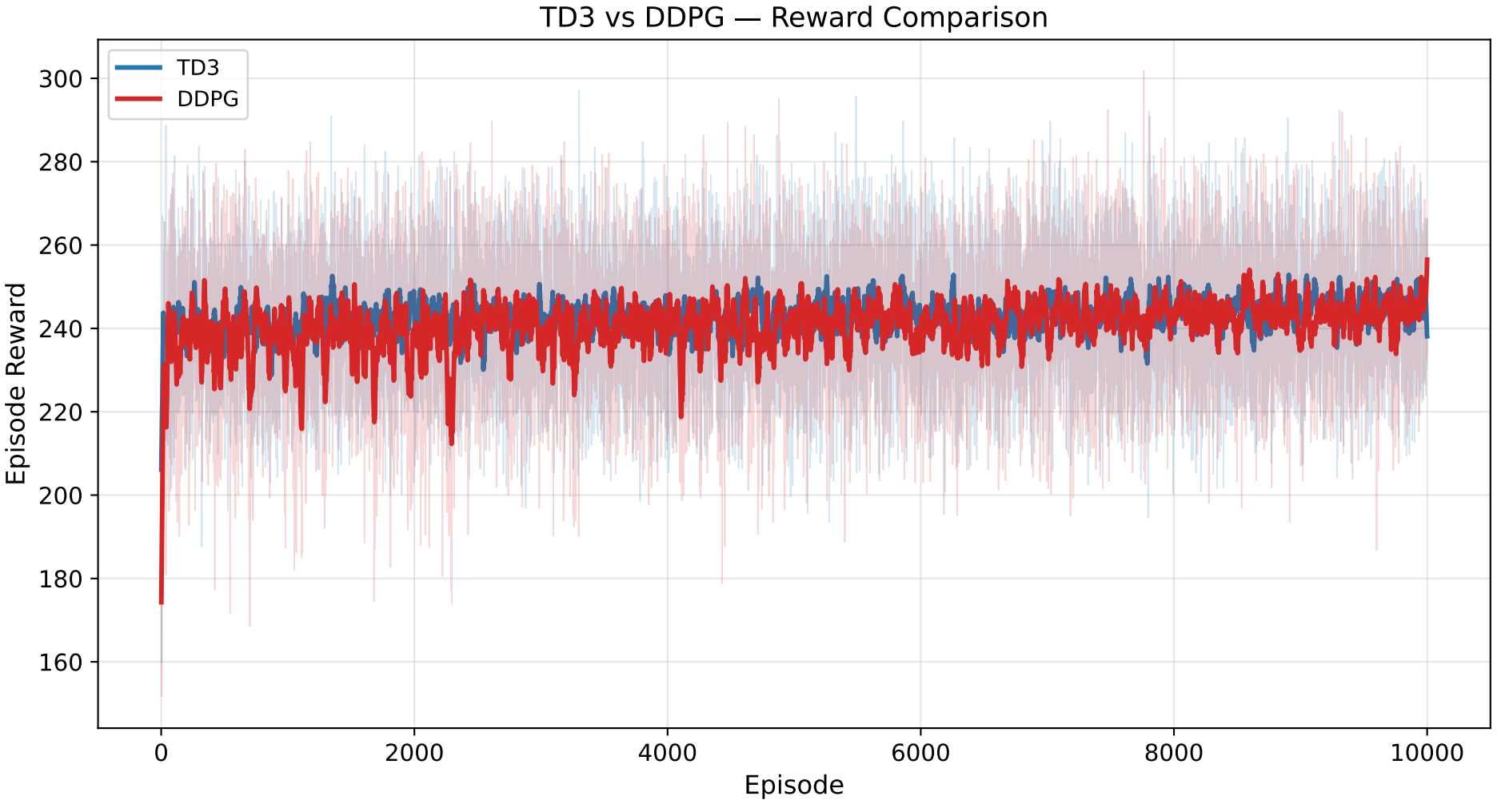

Primary User SINR vs BER — Nakagami-m=3 Fading Channel
(BPSK Modulation | Effect of SU Interference under TD3 vs DDPG)
Bit Error Rate (BER)
PU SINR (dB)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
−5
0
5
10
15
20
25
BPSK Theoretical (AWGN)
BPSK Avg BER (Nakagami-m=3)
TD3 PU (simulated)
TD3 PU Mean BER
DDPG PU (simulated)
DDPG PU Mean BER
PU SINR Threshold (1 linear)


Secondary User (SU) Throughput vs Episodes
TD3 (smoothed)
DDPG (smoothed)
Throughput (bits/s/Hz)
Episode
2.4
2.2
2.0
1.8
1.6
1.4
1.2
0
2000
4000
6000
8000
10000


Primary User (PU) Throughput vs Episodes
TD3 (smoothed)
DDPG (smoothed)
Throughput (bits/s/Hz)
0.9
0.8
0.7
0.6
0.5
0.4
0
2000
4000
6000
8000
10000
Episode


Outage Probability vs Episodes
(SINR_s < 1.0 threshold, rolling per episode)
TD3 (smoothed)
DDPG (smoothed)
5% target
Outage Probability
Episode
0.0
0.2
0.4
0.6
0.8
1.0
0
2000
4000
6000
8000
10000


Episode Reward Curves — TD3 vs DDPG
TD3 Reward Curve
Smoothed reward
Episode Reward
Episode
300
280
260
240
220
200
180
160
0
2000
4000
6000
8000
10000
DDPG Reward Curve
Smoothed reward
Episode Reward
Episode
300
280
260
240
220
200
180
160
0
2000
4000
6000
8000
10000


TD3 vs DDPG — Reward Comparison
TD3
DDPG
Episode Reward
Episode
300
280
260
240
220
200
180
160
0
2000
4000
6000
8000
10000